

LABORATORY PROJECT REPORT

PLC OVER ETHERNET

EXPERIMENT 7

Section 1

Group 2

Semester 2 2025/2026

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Abstract

In this project, an Arduino microcontroller and the OpenPLC Editor are used to construct and test a Programmable Logic Controller (PLC) control system over Ethernet. Developing, simulating, and uploading ladder-diagram logic to operate a basic electromechanical device with push buttons and an LED was the goal. Using common PLC components like contacts, coils, and timer blocks, the experiment started with a simple flashing LED program before moving on to create a Start-Stop control circuit. The ladder logic was built and sent to the microcontroller for real-time execution once the required variables were defined and mapped to the relevant Arduino digital pins. After that, the hardware circuit was built and tested to ensure proper operation. The outcomes validated the usage of PLC-based control using open-source software and microcontroller hardware by showing effective connection between OpenPLC and the Arduino board.

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1.0 Introduction

Programmable Logic Controllers (PLCs) are extensively used in industrial automation due to their reliability, flexibility, and ease of programming, allowing them to control machinery, production processes, and electromechanical systems. With the rise of open-source tools, PLC development can now be integrated with low-cost microcontrollers such as Arduino, giving students the opportunity to explore hands-on automation concepts in a laboratory environment. In this experiment, the OpenPLC Editor is used as the primary software to design ladder logic programs, simulate system behaviour, and interface with an Arduino board that functions as the PLC runtime hardware.

The experiment begins with a simple LED blinking program and advance to a Start-Stop control system, involving the design, simulation, and implementation of ladder-diagram control circuits. To carry out these tasks, students define variables, map physical addresses according to Arduino pin configurations, and upload the developed program via a USB connection. A physical circuit consisting of push buttons, resistors, and an LED is also assembled to test and validate the control logic. Through this exercise, students gain hands-on experience in PLC programming, hardware interfacing, Ethernet/IP-based control concepts, and the integration of microcontrollers with software in basic automation systems.

2.0 Materials and Equipment

2.1 Electronic Components

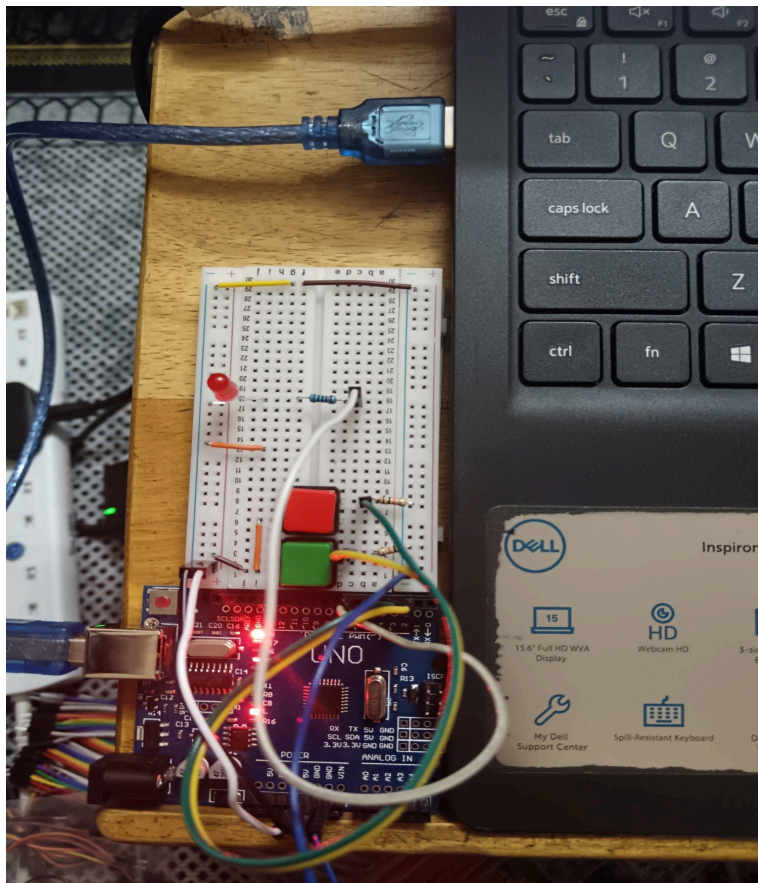
- Arduino Board
- 2 Push Button Switches
- LED
- Resistor
- Jumper Wires

2.2 Equipment and Tools

- OpenPLC Editor Software
- Breadboard

3.0 Experimental Setup

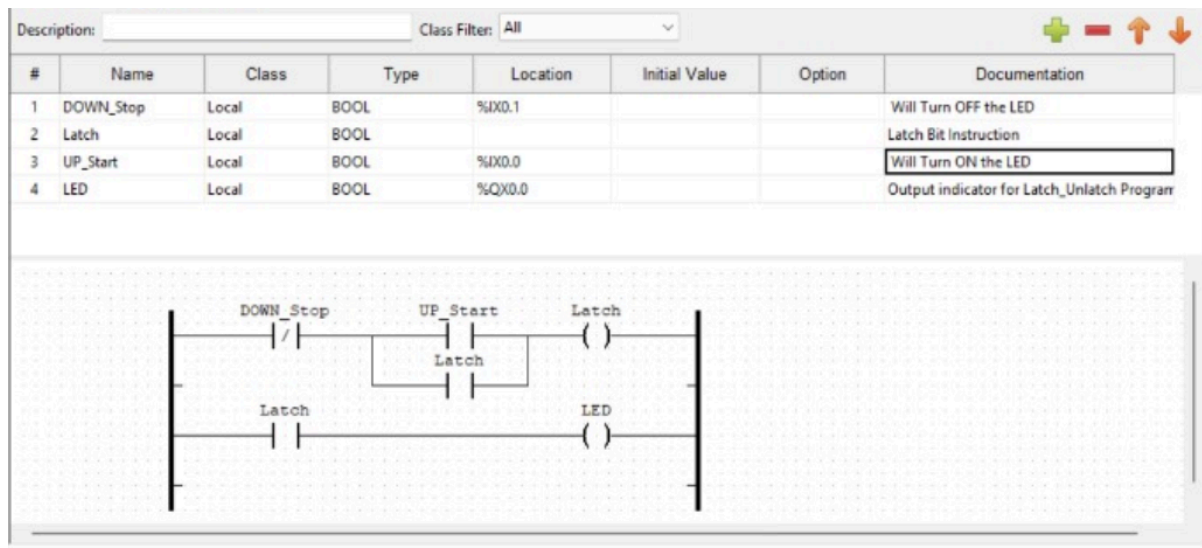
1. Create the ladder diagram.
2. Specify all variables used in the ladder diagram.
3. Compile and simulate the ladder diagram in OpenPLC Editor.
4. Upload the ladder diagram to the Arduino board.
5. Ensure to select correct COM port number and all pin association between the OpenPLC variables and Arduino board.
6. Build the circuit.
7. Test the functionality.



4.0 Methodology

1. Open OpenPLC Editor and create a new ladder diagram project.
2. Declare variables for Start, Stop, and LED output.
3. Draw the Start–Stop ladder diagram using contacts and coil.
4. Compile and simulate the ladder logic to check functionality.
5. Assign each variable to the correct Arduino pin.
6. Connect Arduino to the computer and upload the ladder program.
7. Build the Start–Stop circuit on a breadboard with push buttons and LED.
8. Test the system by pressing Start to turn on the LED and Stop to turn it off.

4.1 Ladder diagram



5.0 Data Collection

During this experiment, data were collected by observing the response of the Start–Stop control circuit when tested with OpenPLC and an Arduino board. The START and STOP buttons were pressed multiple times, and the LED’s behavior was recorded for each trial. Observations were made to confirm whether the LED remained ON after releasing the START button and whether pressing the STOP button successfully turned it OFF. The I/O pin mapping was also checked to ensure that the variables in OpenPLC corresponded correctly to the Arduino pins. In addition, the physical setup, like buttons and wiring was inspected to prevent loose connections or irregular behavior. Overall, the data focused on evaluating the circuit’s responsiveness, stability, and accuracy during operation.

6.0 Data Analysis

Based on the collected data, the performance of the Start–Stop circuit was analysed and found to operate as a typical latching system. Pressing the START button switched the LED ON, and it remained latched due to the holding contact in the ladder logic. When the STOP button was pressed, the latch was immediately broken, turning the LED OFF and confirming correct logic execution. This consistent behavior indicates that the ladder program and pin configurations were properly implemented. Potential issues such as input bouncing or unstable output were also considered, but no significant problems were observed during testing. Overall, the results show that both the software simulation and the hardware implementation aligned with the expected outcomes.

7.0 Results

The experimental results demonstrate that the Start–Stop control circuit operated as intended. The LED turned ON when the START button was pressed and remained ON due to the latching mechanism, while the STOP button consistently switched it OFF. Both the OpenPLC simulation and the physical Arduino implementation produced identical outcomes, confirming that the wiring and programming were accurate. The circuit also maintained consistent performance over multiple test cycles, indicating good stability and reliability. Overall, the objectives of the lab were successfully achieved, and the Start–Stop control concept was effectively implemented.

8.0 Discussion

This experiment successfully demonstrated the operation of a Start–Stop control circuit using OpenPLC and an Arduino platform. Throughout testing, the system performed as expected. Each time the START button was pressed, the LED turned ON and remained ON due to the holding contact implemented in the ladder logic, confirming that the latching concept was correctly applied. The STOP button consistently interrupted the circuit and switched the LED OFF, indicating that the control logic functioned reliably under repeated testing conditions.

This experiment also highlighted the importance of correctly connecting the I/O pins, as established in the weekly module “Start-Stop Control Circuit.” Proper pin configuration ensured that no external factors interfered with the LED’s behaviour. The stable and consistent responses observed during testing indicated that both the programming and wiring were correctly implemented, with no significant issues such as loose connections or timing delays affecting the system performance.

Overall, the results demonstrated a strong correlation between the simulation environment and the physical implementation. This suggests that OpenPLC provides an accurate representation of the system behavior, and that the ladder logic design was both robust and properly implemented.

9.0 Conclusion

In conclusion, the Start–Stop control circuit operated as expected. The LED turned ON when the START button was pressed and remained ON due to the latching logic, while the STOP button consistently switched it OFF. Both the OpenPLC simulation and the physical Arduino setup produced identical results, confirming that the wiring, pin mapping, and ladder program were correctly implemented. This experiment improved our understanding of basic control circuit operation, and all lab objectives were successfully achieved.

10.0 Recommendations

For future improvements, we recommend testing more circuits with components such as relays or motors to see how the Start-Stop logic works in a realistic control system. Also, double-check the wiring connections on the board as there may be things that have been overlooked, such as loose connections and drain connections.

11.0 References

1. Autonomy (Industrial Automation)

<https://autonomylogic.com/>

12. Acknowledgements

Special thanks to Dr. ZULKIFLI BIN ZAINAL ABIDIN and Dr. WAHJU SEDIONO for their guidance and support during this experiment.

Student's Declaration

Certificate of Originality and Authenticity

We hereby certify that we are collectively responsible for the work presented in this report. The content reflects our original work, except where specific references or acknowledgements have been made. No part of this report has been completed or submitted by individuals or sources not identified within.

Furthermore, we confirm that this report is the result of a collaborative effort, with contributions made by all group members. The extent of each individual's contribution is documented within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

Name: AHMAD SYAHMI WAJDI BIN AHMAD SHUKRI	Read	[/]
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Signature: <i>SYAHMI</i>	Agree	[/]
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